

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

ZHIJIE FAN, JI LI, AND DMITRIY ZANIN

ABSTRACT.

CONTENTS

1. Introduction	1
2. Preliminaries	2
3. Analytic tools for Weyl chamber geometry	7
4. Test Section	14
References	24

1. INTRODUCTION

Definition 1.1. Suppose $1 < p < \infty$. We say that a function $f \in L^p_{\text{loc}}(\mathbb{R}^N, dw)$ belongs to Dunkl Besov space $B^p_{\text{Dunkl}}(\mathbb{R}^N)$ if

$$\|f\|_{B^p_{\text{Dunkl}}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \left(\frac{d(x, y)}{\|x - y\|} \right)^p \frac{|f(x) - f(y)|^p}{w(B(x, d(x, y)))^2} dw(x) dw(y) \right)^{1/p} < +\infty.$$

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ -measurable set E by

$$(1.1) \quad (f)_E := \int_E f(x) d\mu(x) := \frac{1}{\mu(E)} \int_E f(x) d\mu(x).$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

Date: August 12, 2026.

2010 Mathematics Subject Classification. 47B10, 42B20, 43A85.

Key words and phrases. Schatten class, Riesz transform commutators, Dunkl operator, Besov space, Nearly weakly orthogonal sequence.

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper. We will denote by $O(x) = \{\sigma(x) : \sigma \in G\}$ the G -orbit of a point $x \in \mathbb{R}^N$, and $O(E) = \bigcup_{x \in E} O(x)$ the G -orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(2.1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2.2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$(2.3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$ as the distance between two G -orbits $O(x)$ and $O(y)$. Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$(2.4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber Γ .

It is clear that $O(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(O(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = O(B(x, r))$.

Denote by $K_{\text{Dunkl},j}(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. *There exists a constant C such that for $j \in \{1, 2, \dots, N\}$ and every x, y with $d(x, y) \neq 0$,*

$$|K_{\text{Dunkl},j}(x, y)| \leq C \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

Lemma 2.3. *For $j \in \{1, 2, \dots, N\}$, and for every ball $B = B(x_0, r) \subset \mathbb{R}^N$, there is another ball $\widehat{B} = B(y_0, r)$ with $y_0 = x_0 + 5re_j$ such that for every $(x, y) \in B \times \widehat{B}$, $K_{\text{Dunkl},j}(x, y)$ does not change sign and satisfies*

$$|K_{\text{Dunkl},j}(x, y)| \geq \frac{C}{w(B(x_0, r))}.$$

Proof. The lower bound is given in [5, Theorem 1.2], while the fact that $K_{\text{Dunkl},j}(x, y)$ does not change sign can be seen in the proof of [5, Theorem 1.2]. Indeed, with the choice $y_0 := x_0 + 5re_j$, we have $y_j - x_j \geq 3r$ for all $(x, y) \in B \times \widehat{B}$. Together with the fact that the heat kernel $h_t(x, y)$ associated with the Dunkl Laplacian is positive, the representation

$$K_{\text{Dunkl},j}(x, y) = -\frac{C}{2} \int_0^\infty \frac{y_j - x_j}{t} h_t(x, y) \frac{dt}{\sqrt{t}},$$

implies that $K_{\text{Dunkl},j}(x, y) < 0$ (and thus, does not change sign) on $B \times \widehat{B}$. □

2.2. Systems of dyadic cubes.

Definition 2.4. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathcal{D} := \bigcup_{k \in \mathbb{Z}} \mathcal{D}_k$, $\mathcal{D}_k := \{Q_\alpha^k : \alpha \in \mathcal{A}_k\}$, of Borel sets $Q_\alpha^k \subseteq X$ is called a *system of dyadic cubes over X* if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha \in \mathcal{A}_k} Q_\alpha^k;$$

(ii) If $\ell \geq k$, then either $Q_\beta^\ell \subseteq Q_\alpha^k$ or $Q_\alpha^k \cap Q_\beta^\ell = \emptyset$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q_\alpha^k \subseteq Q_\beta^\ell$;

(iv) For each (k, α) there exist 2^N indices β such that

$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q \in \mathcal{D}_{k+1} \\ Q \subseteq Q_\alpha^k}} Q;$$

(v) For each (k, α) , one has

$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$

for some positive constants A_1, A_2 depending only on the geometry of X , where we set $B_X(x, r) := B(x, r) \cap X$ for all $x \in X$ and $r > 0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then

$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a *dyadic cube of generation k* , with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.5. For every dyadic cube $Q \in \mathcal{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.6. Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with $(m_1, \dots, m_N) \in \mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber Γ , whose construction will be given in Corollary 3.7.

Lemma 2.7. *There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k \in \mathbb{Z}$, there exists a cube $Q \in \bigcup_{v=1}^{2^N} \mathbb{D}_k^v$ such that*

$$B \subset Q \subset 12 \sqrt{N} B.$$

Proof. • **Step 1.** Recall the one-dimensional setting given in [10, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.6). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \left\{ \left\lfloor \frac{m}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\rfloor, \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\} \mid m \in \mathbb{Z}.$$

These choices together with the nestedness property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

- (1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
- (2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
- (3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a - b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J . Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• **Step 2.** We extend the construction to $\mathbb{R}^N$ ($N \geq 1$).

For $\vartheta = (\vartheta_1, \vartheta_2, \dots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^N I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i -th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \dots, \vartheta_N)$, and let $Q = \prod_{j=1}^N I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N} \ell(Q) \leq 12 \sqrt{N} r_B.$$

This implies that $Q \subset B(x_B, 12 \sqrt{N} r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \dots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.7. $\square$

2.3. Haar Basis. Given a system of dyadic cubes $\mathcal{D}$ over $\mathbb{R}^N$, we now recall the standard way to construct an orthonormal Haar system corresponding to $\mathcal{D}$. For each $k \in \mathbb{Z}$ and $Q \in \mathcal{D}_k$, denote by $Ch(Q) := \{R \in \mathcal{D}_{k+1} : R \subseteq Q\}$. Moreover, we let

$$h_Q^0 := w(Q)^{-1/2} \chi_Q.$$

Fix an enumeration $\{Q_i\}_{i=1}^{2^N}$ of $Ch(Q)$. Let

$$\hat{Q}_i := \bigcup_{j=i}^{2^N} Q_j$$

and

$$h_Q^i := \sqrt{\frac{w(Q_i)w(\hat{Q}_{i+1})}{w(\hat{Q}_i)}} \left(\frac{\chi_{Q_i}}{w(Q_i)} - \frac{\chi_{\hat{Q}_{i+1}}}{w(\hat{Q}_{i+1})} \right), \quad i = 1, 2, \dots, 2^N - 1.$$

Lemma 2.8. *For any $1 < p < \infty$ and $f \in L^p(\mathbb{R}^N, dw)$, we have*

$$f(x) = \sum_{Q \in \mathcal{D}} \sum_{i=1}^{2^N-1} \langle f, h_Q^i \rangle_{L^2(\mathbb{R}^N, dw)} h_Q^i(x),$$

where the sum converges both in the $L^p(\mathbb{R}^N, dw)$ norm and pointwise almost everywhere.

Lemma 2.9. *The family of Haar functions $\{h_Q^i\}_{Q \in \mathcal{D}, 0 \leq i \leq 2^N-1}$ has the following properties:*

- (i) h_Q^i is supported on Q for $i \in \{0, 1, \dots, 2^N - 1\}$;
- (ii) h_Q^i is constant on each $R \in Ch(Q)$ for $i \in \{0, 1, \dots, 2^N - 1\}$;
- (iii) $\int_{\mathbb{R}^N} h_Q^i(x) dw(x) = 0$, for $i \in \{1, 2, \dots, 2^N - 1\}$;
- (iv) $\langle h_Q^i, h_Q^j \rangle_{L^2(\mathbb{R}^N, dw)} = 0$ for $i \neq j$, $i, j \in \{1, 2, \dots, 2^N - 1\}$;
- (v) The collection $\{h_Q^i\}_{i=0}^{2^N-1}$ is an orthogonal basis for the vector space $V(Q)$ of all functions on Q that are constant on each sub-cube $R \in Ch(Q)$.
- (vi) $\|h_Q^i\|_{L^p(\mathbb{R}^N, dw)} \approx w(Q)^{\frac{1}{p}-\frac{1}{2}}$ for $1 \leq p \leq \infty$ and $i \in \{1, 2, \dots, 2^N - 1\}$;

2.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [2–4, 8, 9, 11–14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathcal{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q \in \mathcal{D}}$ is an NWO sequence if

- (1) $\text{supp } e_Q \subseteq cQ$ for some positive constant c independent of Q
- (2) $\|e_Q\|_{L^s(\mathbb{R}^N, dw)} \leq w(Q)^{1/s-1/2}$ for some $s > 2$.

Lemma 2.10. *Let $1 < p < \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. For any compact operator T on $L^2(\mathbb{R}^N, dw)$, we have*

$$\left(\sum_{Q \in \mathcal{D}} |\langle T e_Q, f_Q \rangle_{L^2(\mathbb{R}^N, dw)}|^p \right)^{1/p} \leq C_{N,p,w} \|T\|_{S^p},$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences.

Proof. This property can be found in [12, (1.10)] (see also [8, Section 7]). $\square$

Lemma 2.11. *Let $0 < p < \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that*

$$A = \sum_{Q \in \mathcal{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathcal{D}}$ is a sequence of scalars. We have

$$(2.5) \quad \|A\|_{S^p} \leq C_{N,p,w} \|\{\lambda_Q\}_Q\|_{\ell^p}.$$

Proof. This property can be found in [12, (1.9)] (see also [8, Section 7]). $\square$

3. ANALYTIC TOOLS FOR WEYL CHAMBER GEOMETRY

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \text{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [6, Definition 2.7] *A connected component of*

$$(3.1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

Definition 3.2. [6, Definition 2.9] *A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as*

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. *With the notation given in Definition 3.2, Δ is a basis of W . If we denote by $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$, then the open and closed chambers can be written as*

$$(3.2) \quad \Gamma_+ = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\},$$

$$(3.3) \quad \Gamma = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\},$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+ . By [6, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha \rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \text{span}(\Delta)$ and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion $\text{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\text{span}(\Delta) = W$. On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W , and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [6, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+ , note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ , specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. $\square$

Notation 3.4. With the notation given in Lemma 3.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ .

Lemma 3.5. Each Weyl chamber Γ admits the decomposition

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. $\square$

Corollary 3.6. Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let

$$\begin{aligned} \Phi_W : W &\longrightarrow \mathbb{R}^{N_0}, & \Phi_W(w) &:= (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle), \\ \Phi : \mathbb{R}^N = W \oplus W^\perp &\longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, & \Phi(w + z) &:= (\Phi_W(w), \Phi_{W^\perp}(z)). \end{aligned}$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since Δ is a basis of W , the Gram matrix $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \rightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. $\square$

Corollary 3.7. *Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ .*

Proof. Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}_{\Omega,k}^0$ to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Set

$$\mathbb{D}_\Omega^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that $\mathbb{D}_\Omega^0$ is a system of dyadic cubes over Ω .

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_\Gamma$ is a system of dyadic cubes over Γ . $\square$

Notation 3.8. *Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define*

$$(3.4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

Lemma 3.9. *We have*

$$(3.5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each $v \in R_+$, the coefficients $a_{v,i} \geq 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$ and satisfy $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω ,

$$\int_\Omega f(u, z) d\mu(u, z) = \int_\Gamma f(\Phi(x)) dw(x).$$

We make the linear change of variables $(u, z) = \Phi(x)$, so that $x = \Phi^{-1}(u, z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i \rangle$ for $i = 1, \dots, N_0$. This implies that

$$(3.6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned} \int_{\Omega} f(u, z) d\mu(u, z) &= \int_{\Gamma} f(\Phi(x)) \left(\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx \\ &= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz. \end{aligned}$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. $\square$

Lemma 3.10. *Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and*

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
- (2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2 . By [6, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

- **Step 1.** *Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.*

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1}\sigma_{\beta_{i_k}}w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1}\sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

• **Step 2.** Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \{x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. $\square$

Definition 3.11. We say that $R' \subset \Omega$ is an axis-parallel rectangle if

$$R' := I_1 \times \dots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$(3.7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_\Omega(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\text{dbl}, \Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_\Omega(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_\Omega(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2 \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on L_* and m_* , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We *claim* that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0} \theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c . Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \leq \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4 \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. $\square$

Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$(3.8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$. It follows from (3.8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$(3.9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 3.12, yields

$$(3.10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Lemma 3.13. *For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis-parallel rectangle R' on Ω such that*

$$(3.11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(3.12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis-parallel rectangle

$$(3.13) \quad R' := \left(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. $\square$

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. *There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,*

$$(3.14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

In particular, there exists a constant $C_{\text{dbl}, \Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl}, \Gamma} w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$(3.15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$(3.16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying μ to the inclusions (3.9) and using (3.16) yields

$$(3.17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (3.17) with Lemma 3.12 and the equality (3.6), we obtain that

$$(3.18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (3.14). The second statement follows immediately from (3.14). The third statement follows from the inequalities (3.14) and (2.1). This completes the proof of Theorem 3.14. $\square$

4. TEST SECTION

Lemma 4.1. *Let $K_{\text{Dunkl}}(x, y)$ denote the integral kernel of $(-\Delta_{\text{Dunkl}})^{-1/2}$. Then, for every pair of multi-indices α, β , there exists a constant $C_{\alpha, \beta} > 0$ such that*

$$(4.19) \quad |\partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}}(x, y)| \leq C_{\alpha, \beta} d(x, y)^{-(N-1+|\alpha|+|\beta|)}$$

for all $x, y \in \mathbb{R}^N$ satisfying $d(x, y) > 0$.

Proof. Let $h_t(x, y)$ be the heat kernel of $e^{t\Delta_{\text{Dunkl}}}$. By [1, Theorem 4.1(d)], for every pair of multi-indices α, β , there exist constants $C_{\alpha, \beta}, c_{\alpha, \beta} > 0$ such that

$$(4.20) \quad |\partial_x^\alpha \partial_y^\beta h_t(x, y)| \leq C_{\alpha, \beta} t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} \exp\left(-c_{\alpha, \beta} \frac{d(x, y)^2}{t}\right)$$

for all $t > 0$ and $x, y \in \mathbb{R}^N$.

By (2.1),

$$w(B(z, s)) \gtrsim s^N \prod_{v \in R} (|\langle z, v \rangle| + s)^{k(v)} \geq s^N \prod_{v \in R} s^{k(v)} = s^N.$$

Therefore,

$$V(x, y, \sqrt{t}) = \max\{w(B(x, \sqrt{t})), w(B(y, \sqrt{t}))\} \gtrsim t^{N/2},$$

and hence

$$V(x, y, \sqrt{t})^{-1} \lesssim t^{-N/2}.$$

Substituting this into (4.20), we obtain

$$(4.21) \quad \left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \leq C_{\alpha, \beta} t^{-\frac{N+|\alpha|+|\beta|}{2}} \exp\left(-c_{\alpha, \beta} \frac{d(x, y)^2}{t}\right).$$

By functional calculus,

$$(-\Delta_{\text{Dunkl}})^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{t\Delta_{\text{Dunkl}}} \frac{dt}{\sqrt{t}}.$$

Consequently, away from the orbit diagonal, the corresponding integral kernel is

$$(4.22) \quad K_{\text{Dunkl}}(x, y) = \frac{1}{\sqrt{\pi}} \int_0^\infty h_t(x, y) \frac{dt}{\sqrt{t}}, \quad d(x, y) > 0.$$

Set

$$m := |\alpha| + |\beta|.$$

Using (4.21), we obtain

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}}(x, y) \right| \lesssim \int_0^\infty t^{-\frac{N+m+1}{2}} \exp\left(-c \frac{d(x, y)^2}{t}\right) dt.$$

Therefore

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}}(x, y) \right| \lesssim d(x, y)^{1-N-m} = d(x, y)^{-(N-1+|\alpha|+|\beta|)},$$

which proves (4.19).

The proof is complete. $\square$

Lemma 4.2. *Assume that*

$$\text{span}(R) = \mathbb{R}^N, \quad N > 1, \quad 2 < r < \infty.$$

Let $K_{\text{Dunkl}}(x, y)$ denote the integral kernel of $(-\Delta_{\text{Dunkl}})^{-1/2}$.

Fix a closed Weyl chamber Γ , and let

$$\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$$

be the linear isomorphism appearing in Corollary 3.6. Since $\text{span}(R) = \mathbb{R}^N$, we have

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^N.$$

Let

$$\mathbb{D}_\Gamma = \bigcup_{j \in \mathbb{Z}} \mathbb{D}_{\Gamma, j}$$

be the system of dyadic cubes over Γ constructed in Corollary 3.7. Thus, if $I \in \mathbb{D}_{\Gamma, j}$, then

$$I = \Phi^{-1}(Q)$$

for a unique standard dyadic cube $Q \subset \Omega$ with $\ell(Q) = 2^{-j}$, and we put

$$\ell(I) := \ell(Q).$$

For $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$, let $2Q$ and $3Q$ denote the cubes concentric with Q having side lengths $2\ell(Q)$ and $3\ell(Q)$, respectively, and set

$$I^* := \Phi^{-1}((2Q) \cap \Omega), \quad 3_\Phi I := \Phi^{-1}((3Q) \cap \Omega).$$

Then

$$I \subset I^* \subset 3_\Phi I.$$

There exist nonnegative numbers $\{c_{p,q}\}_{p,q \in \mathbb{Z}^N}$, depending only on N, R, k and the fixed map Φ , such that, for every $M > 0$,

$$(4.23) \quad c_{p,q} \leq C_M(1 + |p| + |q|)^{-M}, \quad p, q \in \mathbb{Z}^N.$$

For every $f \in L^r_{\text{loc}}(\mathbb{R}^N, dw)$ there are functions

$$e_{p,q,I}^{\sigma,\tau}, \quad g_{p,q,I}^{\sigma,\tau}, \quad p, q \in \mathbb{Z}^N, \quad I \in \mathbb{D}_\Gamma, \quad \sigma, \tau \in G,$$

such that, for almost every (x, y) satisfying $d(x, y) > 0$,

$$(4.24) \quad f(x)K_{\text{Dunkl}}(x, y) = \sum_{\sigma, \tau \in G} \sum_{p, q \in \mathbb{Z}^N} c_{p,q} \sum_{I \in \mathbb{D}_\Gamma} \ell(I)^{-(N-1)} w(I)^{1-\frac{1}{r}} \|f\|_{L^r(\sigma I, dw)} e_{p,q,I}^{\sigma,\tau}(x) g_{p,q,I}^{\sigma,\tau}(y).$$

The functions in (4.24) satisfy

$$(4.25) \quad \text{supp } e_{p,q,I}^{\sigma,\tau} \subset \sigma I \subset O(3_\Phi I)$$

and

$$(4.26) \quad \text{supp } g_{p,q,I}^{\sigma,\tau} \subset \tau I^* \subset \tau(3_\Phi I) \subset O(3_\Phi I),$$

$$(4.27) \quad \sup_{\substack{\sigma, \tau \in G \\ p, q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}} w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq 1,$$

$$(4.28) \quad \sup_{\substack{\sigma, \tau \in G \\ p, q \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}} w(I)^{\frac{1}{2}-\frac{1}{r}} \|g_{p,q,I}^{\sigma,\tau}\|_{L^r(\mathbb{R}^N, dw)} \leq 1.$$

Proof. We divide the proof into several steps.

Fix $\sigma, \tau \in G$, and suppose that $x \in \sigma(\Gamma)^\circ$, and $y \in \tau(\Gamma)^\circ$. Set $x_\sigma := \sigma^{-1}x \in \Gamma^\circ$, and $y_\tau := \tau^{-1}y \in \Gamma^\circ$, and $u := \Phi(x_\sigma)$, and $v := \Phi(y_\tau)$.

Since x_σ, y_τ belong to the same closed Weyl chamber, (2.4) gives

$$d(x, y) = d(x_\sigma, y_\tau) = \|x_\sigma - y_\tau\|.$$

Together with (3.8), this yields

$$(4.29) \quad C_\Phi^{-1} \|u - v\| \leq d(x, y) \leq c_\Phi^{-1} \|u - v\|.$$

Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that

$$0 \leq \rho \leq 1, \quad \rho(z) = 1 \quad \text{if } |z|_\infty \leq \frac{1}{4},$$

and

$$\rho(z) = 0 \quad \text{if } |z|_\infty \geq \frac{1}{2}.$$

Define

$$\psi(z) := \rho(z) - \rho(2z).$$

Then

$$(4.30) \quad \text{supp } \psi \subset \left\{ z : \frac{1}{8} \leq |z|_\infty \leq \frac{1}{2} \right\},$$

and, for every $z \neq 0$,

$$(4.31) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

Indeed,

$$\sum_{j=-m}^n \psi(2^j z) = \rho(2^{-m} z) - \rho(2^{n+1} z),$$

and letting $m, n \rightarrow \infty$ proves (4.31).

For $j \in \mathbb{Z}$, and for almost every $u \in \Omega$, there is a unique $Q \in \mathbb{D}_{\Omega, j}^0$ containing u . If

$$I := \Phi^{-1}(Q) \in \mathbb{D}_{\Gamma, j},$$

then $\ell(I) = 2^{-j}$. Therefore, for $x \in \sigma(\Gamma)^\circ$ and $y \in \tau(\Gamma)^\circ$ with $d(x, y) > 0$,

$$(4.32) \quad K_{\text{Dunkl}}(x, y) = \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma, j}} \chi_{\sigma I}(x) K_{\text{Dunkl}}(x, y) \psi \left(\frac{\Phi(\sigma^{-1} x) - \Phi(\tau^{-1} y)}{\ell(I)} \right).$$

For each fixed (x, y) with $d(x, y) > 0$, only finitely many j contribute to the sum, by (4.30).

Write

$$Q = \prod_{i=1}^N [m_i 2^{-j}, (m_i + 1) 2^{-j}], \quad m_i \in \mathbb{Z}_+,$$

and denote its center by c_Q . Suppose that $u \in Q$ and

$$\psi \left(\frac{u - v}{\ell} \right) \neq 0.$$

Then $|u - v|_\infty \leq \frac{\ell}{2}$. Since $|u - c_Q|_\infty \leq \frac{\ell}{2}$, we obtain $|v - c_Q|_\infty \leq \ell$. Hence $v \in (2Q) \cap \Omega$. Equivalently,

$$(4.33) \quad y \in \tau I^*.$$

In particular,

$$\tau I^* \subset \tau(3_\Phi I) \subset O(3_\Phi I).$$

Moreover, by the doubling property, we see that

$$(4.34) \quad w(I) \leq w(I^*) \leq C w(I)$$

uniformly in $I \in \mathbb{D}_\Gamma$.

Fix $\sigma, \tau \in G$ and $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$. Write $\ell := \ell(I)$ and let c_Q be the center of Q .

For

$$a = \frac{u - c_Q}{\ell}, \quad b = \frac{v - c_Q}{\ell},$$

the condition $u \in Q$ means

$$a \in A := [-1/2, 1/2]^N.$$

Furthermore, the condition $v \in (2Q) \cap \Omega$ is equivalent to

$$b \in B_Q := \left\{ b \in [-1, 1]^N : c_Q + \ell b \in \Omega \right\}.$$

Since

$$(c_Q)_i = (m_i + 1/2)\ell, \quad m_i \in \mathbb{Z}_+,$$

we have

$$B_Q = \prod_{i=1}^N [\max\{-1, -m_i - 1/2\}, 1].$$

Thus the left endpoint of the i -th interval is equal to $-1/2$ if $m_i = 0$ and is equal to -1 if $m_i \geq 1$. Consequently, the rectangles $A \times B_Q$ belong to a finite family containing at most 2^N different rectangles.

We claim that there exists a number $\delta > 0$, depending only on N, R, k and Φ , with the following property. For every Q one can choose $\chi_Q \in C_c^\infty((-2, 2)^N \times (-2, 2)^N)$ such that

$$\chi_Q = 1 \quad \text{on } A \times B_Q,$$

$$\text{supp } \chi_Q \subset \{(a, b) : \text{dist}((a, b), A \times B_Q) < \delta\},$$

and, for every integer $M \geq 0$,

$$(4.35) \quad \sup_Q \|\chi_Q\|_{C^M} < \infty.$$

Indeed, there are only finitely many possible rectangles $A \times B_Q$, and hence fixed smooth cutoff functions can be chosen for these finitely many rectangles.

We choose $\delta > 0$ sufficiently small so that the following additional property holds:

$$(4.36) \quad d(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)) \geq c_0\ell$$

whenever $(a, b) \in \text{supp } \chi_Q$ and $a - b \in \text{supp } \psi$. Let us verify this assertion.

Take $(a, b) \in \text{supp } \chi_Q$. There exists $(a_0, b_0) \in A \times B_Q$ such that

$$|a - a_0| + |b - b_0| \leq C_N\delta.$$

If $a - b \in \text{supp } \psi$, then $|a - b|_\infty \geq \frac{1}{8}$. By decreasing δ if necessary, we may assume that $|a_0 - b_0|_\infty \geq \frac{1}{16}$. Set

$$x'_0 = \Phi^{-1}(c_Q + \ell a_0), \quad y'_0 = \Phi^{-1}(c_Q + \ell b_0).$$

Since $c_Q + \ell a_0, c_Q + \ell b_0 \in \Omega$, we have $x'_0, y'_0 \in \Gamma$. Hence (2.4) and (3.8) imply

$$\begin{aligned} d(\sigma x'_0, \tau y'_0) &= d(x'_0, y'_0) \\ &= \|x'_0 - y'_0\| \\ &\geq C_\Phi^{-1}\ell|a_0 - b_0| \\ &\geq c_1\ell. \end{aligned}$$

On the other hand, note that the orbit distance satisfies

$$(4.37) \quad |d(x, y) - d(x', y')| \leq \|x - x'\| + \|y - y'\|.$$

Since Φ^{-1} is linear and bounded,

$$\|\Phi^{-1}(c_Q + \ell a) - \Phi^{-1}(c_Q + \ell a_0)\| \lesssim \ell\delta,$$

and similarly for b, b_0 . Thus (4.37) gives

$$d\left(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)\right) \geq c_1\ell - C\delta\ell.$$

Choosing δ so small that $C\delta \leq c_1/2$ proves (4.36).

Define

$$(4.38) \quad H_{\sigma,\tau,I}(a, b) := \ell^{N-1} \chi_Q(a, b) \psi(a - b) K_{\text{Dunkl}}\left(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)\right).$$

On the support of $\chi_Q(a, b)\psi(a - b)$, (4.36) gives

$$d\left(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)\right) \gtrsim \ell.$$

We claim that, for every integer $M \geq 0$,

$$(4.39) \quad \sup_{\sigma,\tau \in G} \sup_{I \in \mathbb{D}_\Gamma} \|H_{\sigma,\tau,I}\|_{C^M(\mathbb{R}^{2N})} \leq C_M.$$

Indeed, consider a derivative of total order m in (a, b) . Since Φ^{-1} , σ and τ are fixed linear maps, the chain rule give

$$\begin{aligned} & \ell^{N-1} \ell^m \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}}\left(\sigma\Phi^{-1}(c_Q + \ell a), \tau\Phi^{-1}(c_Q + \ell b)\right) \right| \\ & \lesssim \ell^{N-1+m} (c_0\ell)^{-(N-1+m)} \lesssim 1, \end{aligned}$$

where $|\alpha| + |\beta| = m$. Combining this with (4.35) and the fixed smoothness of ψ proves (4.39).

Since $H_{\sigma,\tau,I}$ is supported in the interior of $[-2, 2]^N \times [-2, 2]^N$, its 4-periodic extension in both variables is smooth. Hence it has the Fourier expansion

$$(4.40) \quad H_{\sigma,\tau,I}(a, b) = \sum_{p,q \in \mathbb{Z}^N} b_{p,q,I}^{\sigma,\tau} e^{\frac{\pi i}{2} p \cdot a} e^{\frac{\pi i}{2} q \cdot b},$$

where

$$(4.41) \quad b_{p,q,I}^{\sigma,\tau} = 4^{-2N} \int_{[-2,2]^N} \int_{[-2,2]^N} H_{\sigma,\tau,I}(a, b) e^{-\frac{\pi i}{2} p \cdot a} e^{-\frac{\pi i}{2} q \cdot b} da db.$$

Repeated integration by parts, together with (4.39), gives, for every $M > 0$,

$$(4.42) \quad \sup_{\sigma,\tau \in G} \sup_{I \in \mathbb{D}_\Gamma} |b_{p,q,I}^{\sigma,\tau}| \leq C_M (1 + |p| + |q|)^{-M}.$$

For $a \in A$ and $b \in B_Q$, we have $\chi_Q(a, b) = 1$. Consequently, (4.40) implies

$$\begin{aligned} & K_{\text{Dunkl}}(x, y) \psi\left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)}\right) \\ (4.43) \quad & = \ell(I)^{-(N-1)} \sum_{p,q \in \mathbb{Z}^N} b_{p,q,I}^{\sigma,\tau} \exp\left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q)\right) \exp\left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q)\right) \end{aligned}$$

whenever $x \in \sigma I$, and $y \in \tau I^*$.

The coefficients in (4.43) depend on σ, τ and I . We now absorb this harmless dependence into the normalized rank-one factors.

Define

$$(4.44) \quad c_{p,q} := \sup_{\substack{\sigma,\tau \in G \\ I \in \mathbb{D}_\Gamma}} \left\{ |b_{p,q,I}^{\sigma,\tau}| \left(\frac{w(I^*)}{w(I)} \right)^{1/2} \right\}.$$

By (4.34) and (4.42), for every $M > 0$,

$$c_{p,q} \leq C_M(1 + |p| + |q|)^{-M}.$$

This proves (4.23).

For each σ, τ, p, q, I , define

$$(4.45) \quad a_{p,q,I}^{\sigma,\tau} := \begin{cases} \frac{b_{p,q,I}^{\sigma,\tau}}{c_{p,q}} \left(\frac{w(I^*)}{w(I)} \right)^{1/2}, & c_{p,q} > 0, \\ 0, & c_{p,q} = 0. \end{cases}$$

By definition,

$$(4.46) \quad |a_{p,q,I}^{\sigma,\tau}| \leq 1.$$

Moreover, if $c_{p,q} = 0$, then necessarily $b_{p,q,I}^{\sigma,\tau} = 0$ for every σ, τ, I . Thus, in all cases,

$$(4.47) \quad b_{p,q,I}^{\sigma,\tau} = c_{p,q} a_{p,q,I}^{\sigma,\tau} \left(\frac{w(I)}{w(I^*)} \right)^{1/2}.$$

Fix $\sigma, \tau \in G$ and $I \in \mathbb{D}_\Gamma$. If $\|f\|_{L^r(\sigma I, dw)} > 0$, define

$$(4.48) \quad e_{p,q,I}^{\sigma,\tau}(x) := a_{p,q,I}^{\sigma,\tau} w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{f(x) \chi_{\sigma I}(x)}{\|f\|_{L^r(\sigma I, dw)}} \exp\left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q)\right).$$

If $\|f\|_{L^r(\sigma I, dw)} = 0$, set $e_{p,q,I}^{\sigma,\tau} := 0$. Also define

$$(4.49) \quad g_{p,q,I}^{\sigma,\tau}(y) := w(I^*)^{-1/2} \chi_{\tau I^*}(y) \exp\left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q)\right).$$

The support properties (4.25)–(4.26) are immediate.

Since the Dunkl measure is G -invariant,

$$w(\sigma I) = w(I), \quad w(\tau I^*) = w(I^*).$$

From (4.46) and (4.48), we obtain

$$\|e_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} \leq w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{\|f \chi_{\sigma I}\|_{L^r(dw)}}{\|f\|_{L^r(\sigma I, dw)}} = w(I)^{\frac{1}{r}-\frac{1}{2}}.$$

Therefore,

$$w(I)^{\frac{1}{2}-\frac{1}{r}} \|e_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} \leq 1,$$

which proves (4.27).

Similarly,

$$\|g_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} = w(I^*)^{\frac{1}{r}-\frac{1}{2}}.$$

Hence

$$w(I)^{\frac{1}{2}-\frac{1}{r}} \|g_{p,q,I}^{\sigma,\tau}\|_{L^r(dw)} = \left(\frac{w(I)}{w(I^*)} \right)^{\frac{1}{2}-\frac{1}{r}} \leq 1.$$

This proves (4.28).

Assume first that $\|f\|_{L^r(\sigma I, dw)} > 0$. Combining (4.48), (4.49) and (4.47), we obtain, for $x \in \sigma I$ and $y \in \tau I^*$,

$$c_{p,q} w(I)^{1-\frac{1}{r}} \|f\|_{L^r(\sigma I, dw)} e_{p,q,I}^{\sigma,\tau}(x) g_{p,q,I}^{\sigma,\tau}(y)$$

$$(4.50) \quad = b_{p,q,I}^{\sigma,\tau} f(x) \exp\left(\frac{\pi i}{2\ell(I)} p \cdot (\Phi(\sigma^{-1}x) - c_Q)\right) \exp\left(\frac{\pi i}{2\ell(I)} q \cdot (\Phi(\tau^{-1}y) - c_Q)\right).$$

Indeed, the powers of the measures cancel according to

$$w(I)^{1-\frac{1}{r}} w(I)^{\frac{1}{r}-\frac{1}{2}} w(I^*)^{-1/2} \left(\frac{w(I^*)}{w(I)}\right)^{1/2} = 1.$$

Multiplying (4.43) by $f(x)$ and using (4.50), we get

$$(4.51) \quad \begin{aligned} & f(x) K_{\text{Dunkl}}(x, y) \psi\left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)}\right) \\ &= \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \ell(I)^{-(N-1)} w(I)^{1-\frac{1}{r}} \|f\|_{L^r(\sigma I, dw)} e_{p,q,I}^{\sigma,\tau}(x) g_{p,q,I}^{\sigma,\tau}(y). \end{aligned}$$

If $\|f\|_{L^r(\sigma I, dw)} = 0$, then $f = 0$ almost everywhere on σI , and hence (4.51) continues to hold for almost every $x \in \sigma I$.

Moreover, if $x \in \sigma I$ but $y \notin \tau I^*$, then (4.33) shows that

$$\psi\left(\frac{\Phi(\sigma^{-1}x) - \Phi(\tau^{-1}y)}{\ell(I)}\right) = 0,$$

while $g_{p,q,I}^{\sigma,\tau}(y) = 0$. Thus (4.51) is valid for almost every $(x, y) \in \sigma I \times \tau(\Gamma)$.

We now sum (4.51) over $I \in \mathbb{D}_{\Gamma,j}$ and then over $j \in \mathbb{Z}$. Using (4.32), we obtain, for almost every $x \in \sigma(\Gamma)^\circ$, $y \in \tau(\Gamma)^\circ$, $d(x, y) > 0$, that

$$f(x) K_{\text{Dunkl}}(x, y) = \sum_{p,q \in \mathbb{Z}^N} c_{p,q} \sum_{I \in \mathbb{D}_\Gamma} \ell(I)^{-(N-1)} w(I)^{1-\frac{1}{r}} \|f\|_{L^r(\sigma I, dw)} e_{p,q,I}^{\sigma,\tau}(x) g_{p,q,I}^{\sigma,\tau}(y).$$

Finally, the interiors of the Weyl chambers $\{\sigma(\Gamma) : \sigma \in G\}$ are pairwise disjoint and cover $\mathbb{R}^N$ up to the union of the reflecting hyperplanes, which is a dw -null set. Summing the last identity over $\sigma, \tau \in G$ gives exactly (4.24).

This completes the proof. $\square$

Lemma 4.3. *Suppose that*

$$dw(x) = w(x) dx, \quad w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)},$$

is a nontrivial Dunkl weight, that is,

$$\kappa := \sum_{v \in R} k(v) > 0,$$

and let Γ be a Weyl chamber. Then the following assertion is false in general: if an integral operator V on $L^2(\Gamma, dw)$ has a kernel $K(x, y)$ satisfying

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \leq C_{\alpha,\beta} |x - y|^{-(N-1+|\alpha|+|\beta|)}, \quad x, y \in \Gamma, \quad x \neq y,$$

then

$$\|M_f V\|_{S^{N,\infty}(L^2(\Gamma, dw))} \lesssim \|f\|_{L^N(\Gamma, dx)}$$

for every $f \in L^N(\Gamma, dx)$, with an implicit constant depending only on the constants in the kernel estimates and on the fixed Dunkl data.

More precisely, there exist kernels K_R , with the constants in the above derivative estimates independent of R , and functions $f_R \in C_c^\infty(\Gamma)$ such that

$$\sup_{R \gg 1} \|f_R\|_{L^N(\Gamma, dx)} < \infty,$$

whereas

$$\|M_{f_R} V_R\|_{S^{N, \infty}(L^2(\Gamma, dw))} \longrightarrow \infty \quad \text{as } R \rightarrow \infty.$$

Proof. Choose $\xi \in \Gamma^\circ$. Since ξ lies in the interior of a Weyl chamber,

$$\langle \xi, \nu \rangle \neq 0$$

for every $\nu \in R$. Hence, for every fixed $C > 0$,

$$(4.52) \quad w(R\xi + z) \asymp R^\kappa, \quad |z| \leq C,$$

uniformly for all sufficiently large R . Indeed,

$$\begin{aligned} w(R\xi + z) &= \prod_{\nu \in R} |R\langle \xi, \nu \rangle + \langle z, \nu \rangle|^{k(\nu)} \\ &= R^\kappa \prod_{\nu \in R} |\langle \xi, \nu \rangle + R^{-1}\langle z, \nu \rangle|^{k(\nu)}, \end{aligned}$$

Since $\xi \in \Gamma^\circ$ and the root system R is finite,

$$m_\xi := \min_{\nu \in R} |\langle \xi, \nu \rangle| > 0.$$

Moreover, for $|z| \leq C$,

$$|R^{-1}\langle z, \nu \rangle| \leq R^{-1}C \max_{u \in R} |u|.$$

Thus, for all sufficiently large R , uniformly in $|z| \leq C$ and $\nu \in R$,

$$\frac{1}{2} |\langle \xi, \nu \rangle| \leq |\langle \xi, \nu \rangle + R^{-1}\langle z, \nu \rangle| \leq \frac{3}{2} |\langle \xi, \nu \rangle|.$$

Since $k(\nu) \geq 0$ and R is finite, it follows that

$$0 < c_\xi \leq \prod_{\nu \in R} |\langle \xi, \nu \rangle + R^{-1}\langle z, \nu \rangle|^{k(\nu)} \leq C_\xi < \infty$$

uniformly for $|z| \leq C$ and all sufficiently large R .

Fix nonzero functions

$$\phi, \psi \in C_c^\infty(B(0, \varepsilon)),$$

where $\varepsilon > 0$ is sufficiently small, and choose a fixed vector $a \in \mathbb{R}^N$ such that

$$|a| > 4\varepsilon.$$

Set

$$x_R := R\xi, \quad y_R := R\xi + a.$$

Since $\xi \in \Gamma^\circ$ and a is fixed, for all sufficiently large R ,

$$B(x_R, \varepsilon) \cup B(y_R, \varepsilon) \subset \Gamma^\circ.$$

Define

$$K_R(x, y) := \phi(x - x_R) \overline{\psi(y - y_R)}, \quad x, y \in \Gamma,$$

and let V_R be the corresponding integral operator on $L^2(\Gamma, dw)$:

$$V_R u(x) = \int_{\Gamma} K_R(x, y) u(y) dw(y).$$

For every pair of multi-indices α, β ,

$$\partial_x^\alpha \partial_y^\beta K_R(x, y) = (\partial^\alpha \phi)(x - x_R) \overline{(\partial^\beta \psi)(y - y_R)}.$$

If this expression is nonzero, then

$$|x - x_R| < \varepsilon, \quad |y - y_R| < \varepsilon,$$

and therefore

$$|a| - 2\varepsilon \leq |x - y| \leq |a| + 2\varepsilon.$$

Thus $|x - y| \asymp 1$ on the support of every derivative of K_R , with constants independent of R . It follows that

$$|\partial_x^\alpha \partial_y^\beta K_R(x, y)| \leq C_{\alpha\beta} |x - y|^{-(N-1+|\alpha|+|\beta|)}$$

for all $x \neq y$, where $C_{\alpha\beta}$ is independent of R .

Put

$$\phi_R(x) := \phi(x - x_R), \quad \psi_R(y) := \psi(y - y_R).$$

Then

$$V_R u = \phi_R \langle u, \psi_R \rangle_{L^2(dw)},$$

so V_R is a rank-one operator. Its unique nonzero singular value is

$$s_1(V_R) = \|\phi_R\|_{L^2(dw)} \|\psi_R\|_{L^2(dw)}.$$

By (4.52),

$$\begin{aligned} \|\phi_R\|_{L^2(dw)}^2 &= \int_{\Gamma} |\phi(x - x_R)|^2 w(x) dx \\ &= \int_{\mathbb{R}^N} |\phi(z)|^2 w(R\xi + z) dz \\ &\asymp R^\kappa \int_{\mathbb{R}^N} |\phi(z)|^2 dz, \end{aligned}$$

and hence

$$\|\phi_R\|_{L^2(dw)} \asymp R^{\kappa/2}.$$

Likewise,

$$\|\psi_R\|_{L^2(dw)} \asymp R^{\kappa/2}.$$

Consequently,

$$(4.53) \quad s_1(V_R) \asymp R^\kappa.$$

Choose $\eta \in C_c^\infty(\mathbb{R}^N)$ such that

$$\eta = 1 \quad \text{on } \text{supp } \phi,$$

and define

$$f_R(x) := \eta(x - x_R).$$

For all sufficiently large R ,

$$\text{supp } f_R \subset \Gamma^\circ,$$

and by translation invariance of Lebesgue measure,

$$\|f_R\|_{L^N(\Gamma, dx)} = \|\eta\|_{L^N(\mathbb{R}^N)}.$$

In particular,

$$\sup_{R \gg 1} \|f_R\|_{L^N(\Gamma, dx)} < \infty.$$

Since $f_R = 1$ on $\text{supp } \phi_R$, we have

$$M_{f_R} V_R = V_R.$$

As V_R has rank one,

$$\|V_R\|_{S^{N, \infty}(L^2(\Gamma, dw))} = s_1(V_R).$$

Therefore, by (4.53),

$$\|M_{f_R} V_R\|_{S^{N, \infty}(L^2(\Gamma, dw))} \asymp R^\kappa \longrightarrow \infty,$$

whereas

$$\|f_R\|_{L^N(\Gamma, dx)} = \|\eta\|_{L^N(\mathbb{R}^N)}$$

is independent of R . This proves the assertion. $\square$

REFERENCES

- [1] J. Anker, J. Dziubański, and A. Hejna. Harmonic functions, conjugate harmonic functions and the Hardy space H^1 in the rational Dunkl setting. *J. Fourier Anal. Appl.*, 25(5):2356–2418, 2019. [14](#)
- [2] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [6](#)
- [3] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [6](#)
- [4] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at <https://arxiv.org/abs/2403.08249>, 2024. [6](#)
- [5] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [3](#)
- [6] G. Heckman. Coxeter groups. Lecture notes available at <https://www.math.ru.nl/~heckman/CoxeterGroups.pdf>. Radboud University Nijmegen, 2018. [7](#), [8](#), [10](#)
- [7] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [3](#)
- [8] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [6](#), [7](#)
- [9] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [6](#)
- [10] J. Li, J. Pipher, and L.A. Ward. Dyadic structure theorems for multiparameter function spaces. *Rev. Mat. Iberoam.*, 31(3):767–797, 2015. [5](#)
- [11] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [6](#)
- [12] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon–Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [6](#), [7](#)
- [13] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [6](#)
- [14] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [6](#)

ZHIJIE FAN, SCHOOL OF MATHEMATICAL SCIENCES, SOUTH CHINA NORMAL UNIVERSITY, GUANGZHOU 510631, CHINA
E-mail address: `fanzhj3@mail2.sysu.edu.cn`

Ji LI, DEPARTMENT OF MATHEMATICS, MACQUARIE UNIVERSITY, NSW2109, AUSTRALIA
E-mail address: `ji.li@mq.edu.au`

DMITRIY ZANIN, UNIVERSITY OF NEW SOUTH WALES, KENSINGTON, NSW 2052, AUSTRALIA
E-mail address: `d.zanin@unsw.edu.au`